

Zirong (Mike) Zeng

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EDUCATION

The Chinese University of Hong Kong, Shenzhen <i>B.E. in Computer Science and Engineering</i>	Shenzhen, China Sep 2022 - May 2026
University of California, Berkeley <i>Visiting Undergraduate Student (Berkeley Global Access Program)</i>	Berkeley, CA, USA Jan 2025 - May 2025

SELECTED AWARDS AND HONORS

Competitive Programming

Third Place , 48th International Collegiate Programming Contest (ICPC) Macau Regional (Advanced to 48th ICPC World Finals)	2023
Second Place , 2nd CUHK-Shenzhen Programming Contest	2024
Gold Medal , 47th ICPC Hong Kong, 48th ICPC Jinan, 49th ICPC Kunming Regionals	2023 - 2024
Gold Medal , 9th China Collegiate Programming Contest (CCPC) Shenzhen Regional	2023
Gold Medal , 19th Baidu Star Programming Contest (Collegiate Section)	2023
Silver Medal , National Olympiad in Informatics (NOI)	2021
Silver Medal , Asia-Pacific Informatics Olympiad (APIO)	2020

Academic Scholarships & Honors

School of Data Science (SDS) TA/USTF Award	2024 - 2025
Philip Yu Hong Wong Anita Fung Yee Leung Scholarship for Nobel Class (University's Highest Honor)	2022 - 2025
Dean's List	2022 - 2023 & 2023 - 2024 & 2024 - 2025

PUBLICATIONS

An Extensible and Verifiable Language for Query Rewrite Rules

Sicheng Pan, Shuxian Wang, Wesley Zheng, **Zirong Zeng**, Vijay Sharma, Alvin Cheung
arXiv Preprint 2026

FrontierCS: Evolving Challenges for Evolving Intelligence

Qiuyang Mang, Wenhao Chai, Zhifei Li, ..., **Zirong Zeng**, et al.
International Conference on Machine Learning (ICML) 2026

Electric vehicle routing problem considering traffic conditions and real-time loads

Jingyi Zhao, **Zirong Zeng**, Yang Liu
Transportation Research Part C: Emerging Technologies (TR.C) 2025

RESEARCH EXPERIENCE

University of California, Berkeley <i>Research Assistant (Advised by Prof. Alvin Cheung)</i>	Berkeley, CA May 2025 - May 2026
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- **Project 1: RuleScript**

- Developed a domain-specific language (DSL) for expressing and verifying query optimization rules, which can then be used to generate rule implementations for different database systems.
- Built code generators and implemented 34 rules for Apache Calcite and CockroachDB to demonstrate the practicality of the design.

- **Project 2: FrontierCS**

- Contributed to a coding benchmark for evaluating LLMs on open-ended CS problems; accepted to **ICML 2026**.

Shenzhen Research Institute of Big Data

Research Assistant (Advised by Prof. Xiaodong Luo)

Shenzhen, China

Jun 2023 - Dec 2024

- **Project 1: Linear Programming Solver Optimization**

- Accelerated the Forrest-Tomlin update in the simplex algorithm by identifying and leveraging special permutation properties in matrix structures.
- Achieved **10%-20% speedup** on benchmark LP problems.

- **Project 2: Vehicle Routing Problem (VRP)**

- Explored various local search and large neighborhood search operators to optimize solution quality.
- Contributed to a collaborative effort that achieved **35 new best-known solutions** on the Sartori-Buriol PDPTW benchmark.

INDUSTRY EXPERIENCE

Huawei Technologies

Software Engineering Intern, Container Services

Shenzhen, China

Oct 2024 - Nov 2024

- Contributed to the development of a container image signing plugin using Cosign to strengthen supply chain security in the container deployment pipeline.

SELECTED COURSE PROJECTS

Goat-Wolf Chess

Jan 2023 - May 2023

Developed a strategic board game in Java supporting multiple players; designed interactive GUI with custom graphics, animations, and real-time game state visualization.

Movie Rating Prediction System

Sep 2024 - Dec 2024

Applied Restricted Boltzmann Machines (RBM) for collaborative filtering to predict movie ratings on the Netflix dataset; achieved 2nd place among all teams in class competition as a solo participant.

SERVICES

The Chinese University of Hong Kong, Shenzhen

Undergraduate Student Teaching Fellow (USTF)

Shenzhen, China

Sep 2024 - Dec 2024; Sep 2025 - May 2026

- USTF for **Data Structures**, **Design and Analysis of Algorithms**, and **Discrete Mathematics**, designing homework problems and teaching tutorial sessions for 200+ students.

X-Camp

Instructor for Competitive Programming Courses

Remote

Oct 2025 - Apr 2026

- Taught competitive programming at X-Camp, a Silicon Valley-based coding academy aimed at training top-talent students in computer science.

1st and 2nd Greater Bay Area (GBA) International Programming Contest

Problem Setter

Shenzhen, China

Apr 2025; Apr 2026

1st and 2nd CUHK-Shenzhen Programming Contest

Organizing Committee Member

Shenzhen, China

Apr 2023; Apr 2024